Worksheet for Bayesian Optimization

1. Assume that
$$f \sim GP\left(\mu(x), k(x, y)\right)$$
and that
$$y_i \sim N\left(f(x_i), \sigma^2\right)$$
What is the distribution of $f \mid y_i$? This is well defined even if $\sigma^2 = 0$.

2. How can you represent this in terms of evaluations of the kernel function? If we're doing Bayesian optimization, we may be evaluating the posterior mean and posterior covariance many times for each update.

3. How does this change if we condition on multiple observations? Suppose that we've already conditioned on a set of n observations $\{(x_i, y_i)\}_{i=1}^{n}$. How can we condition on one more?

4. What are the gradients of the posterior mean and variance with respect to x? That is, we assume that
$$f_{post} \sim GP\left(\mu_{post}(x), k_{post}(x, y)\right)$$
In terms of the representation that you found in 2., what is $\nabla_x \mu_{post}(x)$, and $\nabla_x k_{post}(x, x)$?